



KinShield

The Master Build & Business Document

How the product is designed, how it becomes a real app in the market, how every feature works, how it competes against the giants, the full engineering structure, the obstacles in each feature and how to resolve them, the cost and time of building it, and the step-by-step plan to launch it as a startup.

KinShield — the orchestration layer for fraud defence in Canada.

One Trust Score behind every channel · built for the household · twelve languages · honest by engineering discipline.

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How to read this. Every figure is a planning estimate in CAD at realistic 2026 Canadian rates, deliberately conservative. Nothing here claims a capability the operating systems or the law do not permit — that honesty is itself the strategy (see Part 13).

Executive summary

The opportunity. Canadians reported roughly **\$638–643M** in fraud losses to the Canadian Anti-Fraud Centre in 2024 — a near-300% rise since 2020 — and the CAFC estimates only 5–10% of fraud is ever reported, implying true losses in the multiple billions. **Bill C-15** received Royal Assent on 26 March 2026: every Canadian bank must now detect and prevent consumer-targeted fraud, obtain express consent for risky capabilities, let customers set limits, and report fraud data to the FCAC. The regulatory clock has started and it is measured in quarters.

The insight. Every major competitor is an English-first product designed for a tech-comfortable American. The people actually losing the money are **seniors, newcomers and linguistic communities** (CRA, IRCC, job and investment scams in Punjabi, Mandarin, Cantonese, Tagalog, Arabic...), and **small businesses** (deepfake-executive wire fraud). The mismatch between who is targeted and who is served is the founding wedge.

The product. KinShield is not another single-channel detector. It is the **orchestration layer** above the platforms: one **Trust Score** (1–100) behind every channel — texts, links, calls, money, meetings — that knows the **household**, speaks its **twelve languages**, fuses every signal into one judged verdict, decides what to do, and stays with the family through the **aftermath**. Detection is becoming a commodity; **judgment, family context, Canadian context, and recovery are not.**

ALREADY BUILT & WORKING

- A real **Trust Engine** (5 signal families, transparent combiner, 12-language output) running on-device
- A native **iPhone app** (Expo / React Native) — Dashboard, Protect, Map, Family, 30-feature catalogue
- A **web app / PWA** sharing the same engine
- Real camera **QR scanning**, clipboard checking, live community number lookup, calibration reporting

THE ASK

- **Stage 0** validate: \$8–18K, 0–3 months
- **Stage 1** MVP: \$90–160K bootstrap / \$250–350K pre-seed, 3–9 months
- **Stage 2** seed & scale: \$1.5–2.5M, 9–24 months
- **Stage 3** Series A & category: \$8–15M, years 2–4

The one-sentence thesis. Build the P0 wedge for the undefended Canadian household, earn the evidence with published accuracy, and let a 30-feature catalogue compound behind six moats incumbents cannot cheaply copy — while a Bill-C-15 SDK turns the same brain into institutional revenue.

The product & design — and what is already built

2.1 Design language

A restrained, professional dark system — deep ink background, layered surfaces, a single confident emerald accent, and red reserved *strictly* for threats. Vector iconography (no emoji-as-UI), a real typographic scale, the **Trust Score dial** as the product's signature object, and a **Senior Mode** presentation layer (larger type, simplified actions, spoken verdicts) that every feature must render into. Brand promise, printed on the product: *"What's yours, we cannot read. What's the scammer's, we share to protect everyone."*

2.2 Information architecture (the real app)

TAB	PURPOSE	KEY SURFACES
Dashboard	One screen summarising everything	Protection posture ring, 7-day activity, threats-near-you, recent checks, quick actions, alerts
Protect	The front door — Ask KinShield	Paste / Paste-from-clipboard / Scan-QR → on-device verdict + reasoning ledger, spoken aloud
Map	Live threat heat map of Canada	City clusters, category filters, weekly briefing (k-anonymity, city-level only)
Family	The household graph	Members, roles, shared policies, cross-platform alert feed
Features	The A-to-Z catalogue	All 30 features grouped by pillar; Incident Mode, Scam Drill, Settings

2.3 What exists today (this is not a deck — it runs)

- **Trust Engine** — pure-TypeScript, on-device: content / artifact / authenticity / network / anomaly signals, a transparent noisy-OR combiner, a Canadian scam-script corpus (18 named scripts + generic-phishing grammar), lookalike/homoglyph/IP/credential URL forensics, and localized explanations in 12 languages. Unit-tested (25 cases).
- **Native iOS app** (Expo SDK 54) loadable in Expo Go; the same engine powers a **Next.js web app / PWA**.
- **Real on-device capability**: camera QR scanning, clipboard checking, community number lookup, app-wide EN/FR localization, persisted household & history, and a live calibration report.

The honest boundary. Features that need carrier telephony, OS call/SMS entitlements, or partner APIs are present as faithful interactive demos. Part 05 and Part 07 explain exactly how each becomes real.

The market case & sizing

3.1 Verified facts only

- **\$638–643M** reported fraud losses to the CAFC (2024); **~5–10%** reported → true losses in the billions.
- Fraud is the single largest complaint category at Canada's banking ombudsman.
- **INTERPOL (Mar 2026)**: >US\$400B global losses substantially tied to impersonation fraud — the exact AI-voice, spoofed-number attack class KinShield is built against.
- **Bill C-15** (Royal Assent 26 Mar 2026) + the **National Anti-Fraud Strategy** + a new Financial Crimes Agency + Quebec's legislated bank reimbursement = a dated regulatory tailwind.

3.2 Who is losing the money (the segments)

SEGMENT	HOW THEY'RE HIT	WHY KINSHIELD FITS
Seniors	Largest per-incident losses; CRA-arrest, grandparent, tech-support	Senior Mode + Family Circle + the Line + human Recovery
Newcomers / linguistic communities	CRA, IRCC, consulate, job & investment scams in their language	Twelve-language detection & explanations — nobody else does this
Small business	Deepfake-executive wire fraud (the Arup ~US\$25M case)	Wire Guard + Meeting Assist (procedure beats detection)

3.3 Sizing (planning estimate)

	DEFINITION	ESTIMATE
TAM	North-American consumer + SMB fraud-defence & identity protection	US\$10B+ and growing double-digits
SAM	Canadian households + SMBs + the bank/credit-union SDK market under Bill C-15	~\$1.5–2.5B CAD annual
SOM (Yr 3)	~300–500K paid Canadian households + 5–15 institutional SDK customers	\$30–60M CAD ARR path

Bottom-up: 1% of ~15M Canadian households at a \$9.99–19.99 blended ~\$140/yr ≈ \$21M consumer ARR, plus 10 SDK deals at \$50–250K ≈ \$1–2.5M, before insurance and telecom revenue.

Competing with the giants

The rule: name what each competitor is genuinely *best* at — no strawmanning — then state the specific, buildable play that beats them at that exact strength.

COMPETITOR	BEST AT	HOW KINSHIELD WINS
Apple (iOS Call Screening)	Default distribution, free, built-in	Apple commoditizes the receptionist; we sell the bodyguard — judgment, family policy, languages, recovery
Google (Pixel Scam Detection)	Deepest on-device AI	Own the cross-platform household; out-localize; own the after (alert, Incident, human line, insurance)
Norton Genie / Gen Digital	Brand + free AI scam checker	Ask KinShield: proactive, 12-language, Canada-tuned, wired to actions
McAfee Scam Detector	Mass-market suite, 96% claim	Match surface coverage; beat on calibrated honesty & Canadian depth
Truecaller	World's largest caller-ID DB	Judge behaviour not caller-ID; consent-clean network (PIPEDA/Law 25 safe)
Hiya	Carrier/OEM network reach	Take the Canadian carrier road first; own the direct family relationship they never have
Robokiller / YouMail	Answer-bot honeypots	Beat a feature with a system — Line+Honeypot inside a full platform
Aura	Best all-in-one identity (US)	Be Canada's Aura first; add live interception, Senior Mode, pre-evidenced insurance
Incogni / DeleteMe	Data-broker removal	Statutory PIPEDA/Law 25 deletion demands + closed loop into detection
Pindrop / Reality Defender	Enterprise deepfake research	Out-position, don't out-research: multi-signal + procedural guardrails stop a perfect deepfake

4.1 The six moats

#	MOAT	WHY INCUMBENTS CAN'T CHEAPLY COPY IT
1	The Canadian corpus	Labeled scam scripts in 12 languages, built with settlement agencies — data nobody can buy
2	The family graph	Switching costs compound; every protected senior recruits more families
3	Regulatory fluency as product	Bill C-15 kit, FCAC pipeline, Law 25 filings, CRTC/STIR-SHAKEN — years of Canadian legal work
4	Split-plane privacy	E2EE personal data; data-monetizing incumbents can't retrofit it

- | | | |
|---|------------------------|---|
| 5 | The recovery operation | Human, trauma-informed, multilingual — an ops moat, impossible to fake |
| 6 | Calibrated honesty | Published accuracy with failures — competitors' marketing will never allow it |

What we are best at: orchestration (one judged score across every channel), Canadian + multilingual depth where losses concentrate, the household as the unit of protection, owning the aftermath, and a trust posture (published calibration, E2EE) that data-hungry incumbents structurally cannot match.

The features — how each works in the real app

★ = no consumer competitor offers it today. Phase: P0 = MVP (months 0–9) · P1 = scale (9–24) · P2 = years 2–3.

Detect & Decide

Trust Engine & Score ★ P0

Five signal families (content, artifact, authenticity, network, anomaly) fused by a transparent combiner into one 1–100 score + top-3 reasons, localized. The product's vocabulary and the SDK's API contract. **Real mechanism:** distilled on-device models for privacy/latency; GPU server ensembles for heavy authenticity checks; per-language calibration; every decision logged for the quarterly public report.

Ask KinShield ★ P0

Paste, forward, photograph or dictate anything → verdict + action in your language in seconds. **Real mechanism:** share-sheet + keyboard extension + widget entry points; on-device OCR → language ID → engine → action buttons (block/report/alert family/Incident).

SMS / RCS Shield P0

Scam texts filtered before the inbox. **Real mechanism (the key OS detail):** iOS `ILMessageFilterExtension` classifies messages *from unknown senders only* in a sandbox; Android takes the default-SMS or spam role. On-device multilingual classifier + private-relay lookup when confidence is low.

Link & QR Checker P0

Paste or scan; verdict in 2s. **Real mechanism:** domain age / WHOIS / certificate-transparency / homoglyph detection vs a Canadian brand whitelist; in-app camera scanner (built); MV3 browser extension.

VerifyCA ★ P1

Check investment & job offers against the CSA registration search, provincial regulators, and corporate registries. **Real mechanism:** registry scraper/API layer with freshness monitoring + entity resolution (the hard part) + newcomer-language explanations.

Multilingual Defense ★ P0–P1

Twelve languages where losses concentrate. **Real mechanism:** Whisper-class ASR + per-language scam corpora built with settlement-agency partners; a language ships only when paid native speakers validate it.

Zero-Day Anomaly Signals P1

Per-user baselines flag what's abnormal *for you*, catching unlabeled scams. **Real mechanism:** on-device isolation-forest baselines → anomaly score into the Trust Engine; weekly recalibration.

Calls & Voice

The KinShield Line ★ P1

An AI receptionist screens unknown callers and patches through / warns / blocks per family policy. **Real mechanism (off-handset):** conditional call-forwarding of unknown callers to a CPaaS cloud number (Twilio/Telnyx) → streaming ASR → Trust Engine → policy engine → warm transfer with whisper announcement. Handset-independent — works on every iPhone and Android.

Caller Intelligence P0

Label / silence / block flagged numbers before the ring. **Real mechanism:** iOS **Call Directory Extension + Live Caller ID Lookup** (privacy-preserving); Android caller-ID role; STIR/SHAKEN attestation shown as a badge.

Assist Mode ★ P0–P1

Live coaching on a call you answered. **Real mechanism:** user puts the call on speaker; the app analyses ambient audio via the **microphone** (permitted) entirely on-device — lawful under one-party consent; nothing leaves the phone.

Voice Lock P1

Family voiceprints verify in-app requests. **Real mechanism:** ECAPA-TDNN-class embeddings on-device (never raw audio); liveness challenge + anti-replay; Quebec CAI biometric declaration before launch there.

Honeytrap Decoy Line ★ P2

Decoy AI personas waste scammer time & harvest scripts. **Real mechanism:** CPaaS decoy numbers + persona library on our own consent/recording footing; harvested intel feeds the corpus. Launches only behind counsel review (interception/CRTC); never uses a user's voice.

Family & Seniors

Family Circle ★ P0

The household is the unit of protection — up to 12 members, cross-platform alerts, policy engine, the senior never manages a password (family-share key recovery). **Real mechanism:** household graph + role/permission model consumed by every shield; audited approval log on the personal plane.

Senior Mode ★ · Scam Drill ★ · Briefing & Heat Map P0–P1

An alternate presentation layer every feature renders into; opt-in scam inoculation drills with a household scoreboard; a localized weekly threat briefing + live map (city-level, k-anonymity).

Money & Business

Check Before You Send ★ · Wire Guard ★ · Meeting Assist ★ P1–P2

The 10-second habit before money moves (recipient + three pressure questions + cooling-off); the SMB Arup-killer (out-of-band release ceremony: live selfie + spoken codeword + voice match — the deepfake can't pass the channel it doesn't control); a second opinion on high-stakes video calls (advisory flags + the procedural guardrail).

KinShield SDK ★ (the Bill C-15 product) · Insurance Benefit P1-P2

The same brain licensed to banks & credit unions: Trust Engine API + a C-15 control kit + threat-intel feed + FCAC reporting pipeline. A fraud-reimbursement benefit with a Canadian insurer, claims pre-evidenced by Incident Mode.

Privacy, Exposure & Recovery

Zero-Knowledge Vault ★ · Exposure Sweep · Passkey Coach P0-P1

E2EE personal data verified by a published audit; breach + data-broker sweeps with statutory Law-25 deletion demands and Shadow Contacts; a gamified household hardening program.

Incident Mode ★ · Crisis Vault ★ · Recovery Line ★ · Long-Con Radar ★ P0-P1

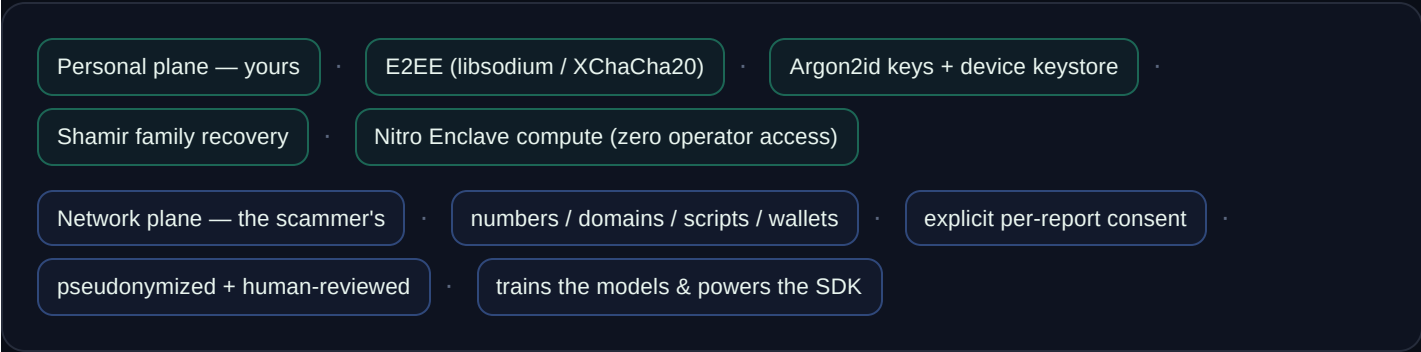
The panic button (guided first-hours flow, pre-filled CAFC/police, scripted bank calls, recovery-scam pre-arm); the sextortion first-four-hours flow (de-escalation + StopNCII); a trained multilingual human who joins bank calls for 3 months; staged detection of romance/pig-butchering over weeks.

Community Threat Network · Email Guard P0-P1

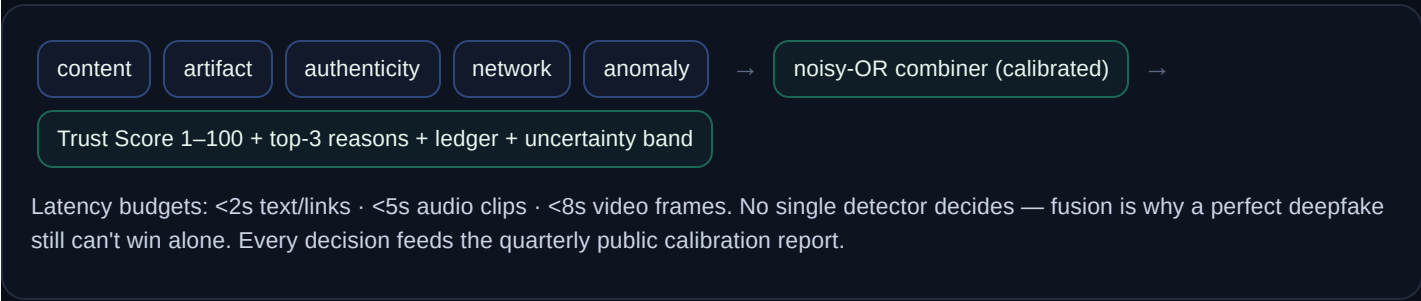
One consented report protects everyone in minutes (graph propagation, reputation decay, contributor karma); read-only OAuth inbox scanning with SPF/DKIM/DMARC forensics.

The engineering structure

6.1 The split data-plane (the architectural promise)



6.2 The Trust Engine



6.3 The stack

LAYER	CHOICE	RATIONALE
Mobile	Native Swift (iOS) + Kotlin (Android); RN for shared surfaces	OS extensions (SMS Filter, Call Directory, Live Caller ID, Android roles) are unreliable from cross-platform frameworks
Backend	Go + TypeScript services; Postgres (+pgvector, graph); Redis; NATS/Kafka; AWS ca-central-1	Boring, hireable, auditable; Canadian residency is a sales feature
Telephony	CPaaS (Telnyx/Twilio), Canadian DIDs, media-stream forking, dual-provider failover	The Line & Honeypot live or die on telephony reliability
ML	PyTorch training; Core ML / TF Lite on-device; L4-class GPU inference; labeled-corpus pipeline as first-class infra	On-device where privacy demands, server where accuracy demands — the corpus pipeline IS the company
Security & compliance	SOC 2 Type I → II; annual pen test; crypto audit; PIPEDA PIAs; Law 25 / CAI declarations; OSFI B-13 mapping	Compliance artifacts double as B2B sales collateral

6.4 Infrastructure economics (monthly opex, CAD, planning ranges)

COST LINE	10K USERS	100K USERS	500K USERS
Cloud compute & storage	\$1.5–3K	\$8–16K	\$30–60K
GPU inference	\$0.8–2K	\$6–14K	\$25–55K
Telephony (Line + Honeypot, ~15% active)	\$0.5–1.5K	\$8–20K	\$45–110K
Third-party APIs (breach, crypto-intel, WHOIS/CT, registries)	\$1–2K	\$5–10K	\$15–30K
Observability & security tooling	\$0.5–1K	\$2–4K	\$6–12K
TOTAL INFRA	≈\$4–10K	≈\$29–64K	≈\$121–267K

Read against \$9.99–19.99 pricing: 75–85% gross margins at scale are realistic if the human Recovery Line is gated to paid tiers and telephony abuse is controlled. The single biggest cost lever is **Line minutes** — which is why screening defaults to "unknown callers only."

Obstacles in every feature — and how each is resolved

FEATURE	THE REAL OBSTACLE	THE RESOLUTION
SMS / RCS Shield	iOS exposes only unknown-sender messages in a sandbox with no free network; Android needs a Play "Permissions Declaration" (rejection risk)	On-device distilled classifier (no network needed); private-relay deferred lookup; ship a clear demo video + justification for the declaration — Truecaller/Hiya pass this routinely
Ask KinShield	Multimodal OCR + 12-language inference cost and latency	On-device OCR first; staged languages (EN/FR/PA/ZH at P0, +2/quarter); cache verdicts per thread
Link & QR Checker	Real-time domain intelligence is rate-limited and costly; sandboxed render is heavy	Cache + CT-log streaming; lexical forensics (homoglyph/IP/lookalike) need no network and catch most spoofs (already built)
The KinShield Line	Per-minute telephony is the top opex lever ; conditional-forwarding setup differs per carrier; latency on warm-transfer	Default to unknown-callers-only; per-account minute budgets; dual-CPaaS failover; a per-carrier setup wizard; streaming ASR to keep transfer <2s
Caller Intelligence	Apple gates the Live Caller ID entitlement (weeks of review); blocklists go stale (scammers burn numbers daily)	File the entitlement in week one; bloom-filter local cache + private lookup; lean on behaviour (the Line) not just number identity
Assist Mode	On-device ASR battery/thermal budget; accuracy on noisy speakerphone audio	Distilled streaming ASR; thermal/battery caps; speaker-focus; degrade gracefully to keyword spotting
Voice Lock	Biometric law (Quebec CAI, BIPA-style liability); spoofing/replay; senior enrollment friction	File CAI declaration before Quebec; store embeddings not audio; liveness + anti-replay; quarterly re-enrollment; senior-designed UX
Honeypot	Interception/recording & CRTC law; reputational risk; never a user's cloned voice	Counsel gate; our own DIDs & consent footing; strict multi-report targeting thresholds (precedent: O2 "Daisy", Apate.ai)
Family Circle	Monitoring a parent raises consent & coercive-control risk	Explicit consent ceremonies; transparency to the protected member; no covert monitoring; auditable approval log
Incident Mode	No public APIs freeze accounts or file police/CAFC reports automatically	Pre-filled deep links + sequenced playbooks maintained by content ops; automation deepens as bank SDK relationships land
Crisis Vault	Safeguarding, minors, mental-health liability	Scripts vetted with mental-health partners; age-appropriate gating; NCMEC TakeItDown / StopNCII; mandatory safeguarding review
Recovery Line	Hiring/training trauma-informed multilingual staff; ops cost; advice liability	QA'd scripts; strict scope (navigation, not legal advice); gated to paid tiers; staffing scales with revenue

Insurance Benefit	12+ month underwriter negotiation; actuarial risk; OSFI	Partner an established insurer/MGA; realistic caps (\$10–25K); pre-evidenced claims; "designed fast", never "instant"
Exposure Sweep	Broker compliance at scale; breach-data licensing	Legal-demand templating engine; HIBP-class licensing; statutory PIPEDA/Law 25 teeth competitors don't use
VerifyCA	13 provincial registries, no official APIs, entity resolution ("MapleWealth Capital Inc." across registries)	Scraper layer with freshness monitoring + fuzzy entity resolution + OCR for image ads
Wire Guard / Meeting Assist	SMB integration; screen-capture consent; deepfake detectors generalize poorly	Procedural out-of-band release (beats detection); webhook integrations (QuickBooks/Stripe); advisory flags, never verdicts
KinShield SDK	6–12 month bank sales cycles; SOC 2 prerequisite; integration lift	Credit unions & digital banks first; compliance artifact pack; Big-5 in Year 3 on live references
Trust Engine / ML	Scarce Canadian multilingual scam data; model drift; adversarial evasion; honest calibration	The corpus pipeline + community + honeypot data flywheel; drift monitoring; published quarterly calibration (the moat)
Zero-Knowledge Vault	E2EE vs recovery tension (senior lockout); audit cost	Shamir family key-shares; external crypto audit (~\$20–30K) published before the claim is marketed

Cross-cutting risks — the company-level obstacles

RISK	RESOLUTION
App Store / Play policy — extensions & SMS/call roles can be rejected or rules can change	Use only sanctioned APIs; file entitlements early; maintain a demo build & justification; keep the network plane consent-clean; precedent exists (Truecaller, Hiya, Robokiller, Norton all ship these)
Unit economics — Line minutes can erode the \$9.99 tier	Unknown-callers-only default; minute budgets; gate the Line/Recovery to paid tiers; model at \$0.015/min blended
CAC / retention — consumer security churns	Partnership-sourced CAC <\$20 (settlement agencies, seniors' orgs); the family graph drives retention & virality
Fraudsters adapt	Anomaly net for zero-days; the honeypot/community flywheel; ship calibration so adaptation is visible, not hidden
Privacy regulators (PIPEDA, Law 25, CAI)	Compliance is the product (Moat 3); PIAs, declarations, residency — turned into the SDK
Trust is fragile — one overclaim detonates the brand	Claims discipline (Part 13): never publish a number without its benchmark, date & failures
Talent & key-person — telephony + ML + native are rare skills	Hybrid build (founder + 2 specialist contractors at MVP); boring/hireable backend; IP-assignment from commit one
Capital intensity	Phase gates stop Stage-2 spend on a Stage-0 idea; SR&ED returns ~35% of eligible R&D; IRAP co-funds technical work

Cost & time

9.1 Per-feature build cost (one-time engineering, CAD, realistic ranges)

FEATURE	COST	BUILD	PHASE
Trust Engine & Score	\$70–100K	8–10 wk	P0
Ask KinShield	\$45–65K	6–8 wk	P0
SMS / RCS Shield	\$40–60K	6–8 wk	P0
Link & QR Checker	\$30–45K	4–6 wk	P0
Caller Intelligence	\$35–55K	6–8 wk	P0
Family Circle (+ Senior Mode)	\$60–85K	10–12 wk	P0
Community Threat Network	\$45–70K	8–10 wk	P0
Incident Mode	\$50–75K	8–10 wk	P0
Zero-Knowledge Vault (+ audit)	\$70–100K	12–14 wk	P0
Multilingual (to 12)	\$90–150K	staged	P0–P1
The KinShield Line	\$130–190K	14–18 wk	P1
Assist Mode	\$55–80K	8–10 wk	P0–P1
Check Before You Send	\$30–45K	5–7 wk	P1
Crisis Vault	\$50–75K	9–12 wk	P1
Email Guard	\$55–80K	8–10 wk	P1
Exposure Sweep	\$60–90K	10–12 wk	P1
Long-Con Radar	\$60–85K	10–12 wk	P1
VerifyCA	\$45–65K	6–8 wk	P1
Voice Lock	\$40–60K	7–9 wk	P1
Zero-Day Signals · Passkey · Scam Drill · Briefing	\$125–195K	—	P1
Recovery Line (setup)	\$30K + \$15–30K/mo	by mo 12–15	P1
KinShield SDK (to pilot)	\$140–200K	16–20 wk	P1 → P2
Honeypot · Insurance · Meeting Assist · Wire Guard	\$320–490K	—	P2

9.2 Phase roll-up & funding

PHASE	BUILD COST	FUNDING	TIME
Stage 0 · Validate	—	\$8–18K	0–3 mo
Stage 1 · MVP (P0)	\$445–650K full / \$250–350K bootstrap cut	\$90–160K bootstrap / \$250–350K pre-seed (SAFEs)	3–9 mo
Stage 2 · Seed & scale (P1)	\$840K–1.23M + Recovery opex	\$1.5–2.5M seed (cap \$8–14M)	9–24 mo
Stage 3 · Series A (P2)	\$420–620K + partnerships	\$8–15M Series A	yr 2–4
TOTAL TO FULL CATALOGUE	≈\$1.7–2.5M OVER ~3 YRS	SR&ED returns ~⅓ of eligible R&D; inside the seed+A envelope	

9.3 Pricing & revenue logic

TIER	PRICE (CAD)	ROLE
Free	\$0	Distribution + network data (SMS Shield, basic checks, Incident self-serve)
Family	\$9.99/mo	The wedge — bought by adult children (Circle, Line, Crisis Vault, Long-Con, Vault)
Premium	\$19.99/mo	Margin tier (Recovery Line, Wire Guard, Voice Lock, Insurance)
Enterprise SDK	\$50–250K/yr + usage	The institutional revenue engine (Trust API, C-15 kit, intel feed, FCAC pipeline)

The phased roadmap — step by step

Each stage has an entry condition, a budget, a task list, and an **exit gate**. Do not start a stage until the previous gate is passed — the gates stop you spending Stage-2 money on a Stage-0 idea.

Stage 0 — Validate (0–3 mo · \$8–18K)

Steps: clear the name (CIPO + domains + collisions); incorporate federally + founder IP assignment; **100 discovery interviews** (25 seniors, 25 adult children, 25 newcomers, 15 victims, 10 bank/CU risk staff); landing page + waitlist per language with \$1–2K ads; a **concierge "no-code" MVP** (you manually check suspicious texts for 50 families for 4 weeks — builds the first labeled corpus by hand); one settlement-agency/seniors-org LOI; 2–3h with a privacy/fintech lawyer; apply for IRAP/SR&ED/Mitacs.

Challenge → **resolution:** a deck feels like progress while proving nothing → force the concierge test & interviews; if <15% call the problem serious or the concierge test doesn't get repeat sends, **pivot the wedge for \$5K**, don't build.

Exit gate: ≥1,500 waitlist · 1 LOI · concierge users send suspicious content more than once · name cleared.

Stage 1 — MVP (3–9 mo · \$90–350K)

Scope (zero app-store risk): SMS/RCS Shield, Link & QR, Ask KinShield, Trust Score v1 (4 languages), Family Circle + Senior Mode, Incident Mode, Community reporting. **No** Line, insurance, or SDK yet — those need the seed.

Team (hybrid): founder + 1 senior mobile contractor + 1 ML contractor + paid community validators.

Challenge → **resolution:** native message/call hooks & the first ML model are the two hard, narrow pieces → hire specialist contractors only for those; App Store/Play review for the SMS extension → prepare the sensitive-permission declaration + demo video early (month 3); Apple Live Caller ID entitlement queue → file in week one.

Exit gate: 10K installs · 500+ paying households (~\$60K ARR) · D30 retention ≥25% · one documented, consented "money saved" story · partnership-sourced CAC <\$20.

Stage 2 — Seed & scale (9–24 mo · \$1.5–2.5M)

Ships, in order: call layers (Caller Intel, Assist, voicemail) + Exposure Sweep → the KinShield Line (limited → GA) + VerifyCA + Check Before You Send → Voice Lock (Quebec after CAI) + Crisis Vault + **SDK pilot #1 with a credit union** → Wire Guard + languages 9–12 + SDK pilot #2 + honeypot legal review + insurance-partner negotiations begin. Hire (in order): senior backend/telephony, ML, mobile, head of community partnerships, part-time compliance, 2 recovery staff.

Challenge → **resolution:** telephony cost & reliability → dual-CPaaS, minute budgets, paid-tier gating; SOC 2 is a sales prerequisite → start Type I in month 9; insurance underwriting is a 12-month pole → begin negotiations in P1.

Exit gate: \$1M+ ARR · 2 SDK pilots signed · churn <3.5%/mo · SOC 2 Type I.

Stage 3 — Series A & category (yr 2–4 · \$8–15M)

Goal: bank SDK revenue, the insurance benefit live, market leadership. Honeypot & Meeting Assist ship; SDK goes multi-tenant production; Big-5 / credit-union-central conversations on the back of SOC 2 Type II + live references.

Challenge → **resolution:** enterprise procurement & security review → the compliance artifact pack built as P1/P2 collateral; scaling the human Recovery Line → train-the-trainer + multilingual partner network.

Exit gate: \$5–10M ARR · a Big-5 or credit-union-central deal · path to profitability.

What founders forget — added here

OFTEN FORGOTTEN	WHY IT MATTERS / WHAT TO DO
Native extension review lead time	SMS Filter, Call Directory & Live Caller ID are separate iOS targets with their own review + entitlements — file in week one, not at launch
Content operations	Incident playbooks, bank phone scripts & registry lists rot — budget a content-ops function, not just engineering
Abuse & cost controls on the Line	A single abused account can blow the telephony budget — minute caps, rate limits, fraud-on-the-fraud-tool monitoring from day one
Accessibility as acceptance criterion	"If it doesn't work for a 78-year-old in Punjabi, it isn't done" — Senior Mode is a gate on every feature, validated in paid usability studies
Data poisoning & model drift	Contributor karma + human review at ingestion; drift monitoring + per-language calibration curves
Coercive-control / consent edge cases	Family monitoring can be abused — explicit consent ceremonies, transparency to the protected member, no covert mode
Support & recovery as a real org	Trauma-informed, multilingual, QA'd — a hiring & training pipeline, not a ticket queue; it's also the source of the brand's stories
Unit economics per cohort	Track infra-per-paying-household (\$1–4 at scale) and Line-minute cost per cohort, not just blended margin
Trademark, entity, cap table hygiene	CIPO/NUANS clearance, federal incorporation, founder IP assignment & SAFEs done cleanly before any marketing
Non-dilutive capital	SR&ED (~35% of eligible R&D back) and IRAP can mean the difference between 6 and 10 months of runway — file from the first line of R&D code
Deepfake-detector limits (don't oversell)	Detectors generalize poorly to unseen synthesis — rely on multi-signal fusion + procedural guardrails (Wire Guard), and say so
Crisis safeguarding & minors	Sextortion flows touch minors and mental health — partner review and safeguarding sign-off are mandatory gates, not nice-to-haves

The numbers that matter — KPIs by stage

STAGE	PRIMARY KPIS	TARGET
0 · Validate	Waitlist; LOIs; concierge repeat-send rate; problem-severity %	1,500 / 1 / repeat >1 / ≥15%
1 · MVP	Installs; paying households; D30 retention; documented saves; CAC	10K / 500+ / ≥25% / ≥1 / <\$20
2 · Scale	ARR; SDK pilots; monthly churn; SOC 2; languages live; Line minutes/user	\$1M+ / 2 / <3.5% / Type I / 8–12 / controlled
3 · Category	ARR; institutional deal; gross margin; calibration precision/recall (published)	\$5–10M / 1 Big-5 or CU-central / 75–85% / published quarterly

Calibrated honesty means one of the headline KPIs is **published accuracy with its failures** — precision, recall, F1 and the misses, dated, per language. It is computed live in the product today (the Calibration page).

Claims discipline & ethics — the moat that costs nothing

The permanent rules — the integrity standard that is itself a competitive weapon:

- Never publish an accuracy number without the benchmark, the date and the failure cases beside it.
- Never imply protection on a channel the OS doesn't permit (the Line screens what's routed to it; Assist helps when invoked — and the marketing says exactly that).
- Never attribute a breach or failure to a competitor without primary-source verification.
- Never promise insurance outcomes — "designed to be fast", never "instant" or "no proof of loss".
- Never use invented pilot data; never collect what the personal plane cannot protect; never ship a language whose detection quality isn't validated by paid native speakers.

In a market built on "96%" and "99.2%", the company that publishes its calibration curves — misses included — becomes the one journalists cite, regulators trust, banks shortlist, and families recommend. That is the genius this product runs on: not more claims than the giants, but claims that are **true**.

What exists today & how to run it

A.1 Repository

PATH	WHAT
mobile/	Native iPhone app (Expo SDK 54, React Native) — runs in Expo Go; the flagship surface
mobile/lib/trust-engine	The on-device Trust Engine (shared, pure TypeScript)
src/ (repo root)	Next.js web app / PWA sharing the same engine
src/lib/trust-engine	Engine source of truth + 25 unit tests + live calibration harness

A.2 Run it

iPhone (Expo Go): `cd mobile && npm install && npx expo start` → scan the QR with the iPhone Camera (same Wi-Fi). See `mobile/HOW-T0-RUN-IPHONE.md`.
Standalone build (EAS): `eas build --profile preview --platform ios`. See `mobile/BUILD-STANDALONE.md`.
Web: `npm install && npm run dev` → `localhost:3000`. See `HOW-T0-RUN.md`.

A.3 The path to a real, shipping app — the native extensions to add on the EAS build

CAPABILITY	IOS	ANDROID
Smishing defence	ILMessageFilterExtension (unknown senders)	Default-SMS / spam role + Permissions Declaration
Caller defence (pre-ring)	Call Directory + Live Caller ID Lookup	Caller-ID & spam role; CallScreeningService
Call screening (the Line)	Network layer — CPaaS conditional call-forwarding (handset-independent)	
Live call coaching	Assist Mode via microphone, on-device (one-party consent)	

The summary. Apple/Android don't forbid the outcomes — they channel them through sanctioned APIs (unknown-sender SMS filtering, Call Directory/Live Caller ID) and, for the hardest part, you move screening to the network layer where you're allowed. Every KinShield capability has a legitimate, approvable path; the value was never privileged device access — it's judgment, family, Canadian + multilingual context, and recovery.